



ΣΧΟΛΗ ΨΗΦΙΑΚΗΣ ΤΕΧΝΟΛΟΓΙΑΣ ΤΜΗΜΑ ΠΛΗΡΟΦΟΡΙΚΗΣ ΚΑΙ
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3^η ΟΜΑΔΑ ΑΣΚΗΣΕΩΝ

ΓΙΑ ΤΟ ΜΑΘΗΜΑ

"Θέματα Σχεδίασης Βάσεων Δεδομένων"

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1ο μέρος:Βασικό σενάριο και οντότητες

(3h_ergasia.sql file)

Table book

```
--BOOKS relationship
CREATE TABLE BOOKS (
    TITLE VARCHAR2(60),
    WRITER VARCHAR2(60),
    ISBN NUMBER(10),
    CATEGORY VARCHAR2(60)
);
```

```
ALTER TABLE BOOKS
ADD CONSTRAINT PK PRIMARY KEY(ISBN);

SELECT * FROM BOOKS;
DESC BOOKS;
```

Table members

```
--MEMBERS relationship
CREATE TABLE MEMBERS(
    ID NUMBER(5) ,
    NAME VARCHAR2(60),
    ADDRESS VARCHAR(60)
);

-- A sequence in order to do auto increment for the id field
CREATE SEQUENCE members_seq
    MINVALUE 1
    START WITH 1
    INCREMENT BY 1;

ALTER TABLE MEMBERS
ADD CONSTRAINT PKM PRIMARY KEY(ID);

SELECT * FROM MEMBERS;
DESC MEMBERS;
```

Table borrows

```
--BORROW relationship
CREATE TABLE BORROWS(
    BORROWID NUMBER(5) ,
    BOOKISBN NUMBER(10),
    BORROWDATE DATE,
    RETURNDATE DATE,
    AVALIABILLITY NUMBER(1),
    MEMBERID NUMBER(5)
);

--A sequence in order to do auto increment for the field BORROWID
CREATE SEQUENCE borrows_seq
    MINVALUE 1
    START WITH 1
    INCREMENT BY 1;
```

```

ALTER TABLE BORROWS
ADD CONSTRAINT PKB PRIMARY KEY(BORROWID);

ALTER TABLE BORROWS
ADD CONSTRAINT FK_MID FOREIGN KEY(MEMBERID) REFERENCES MEMBERS(ID);

ALTER TABLE BORROWS
ADD CONSTRAINT FK_BOOK FOREIGN KEY(BOOKISBN) REFERENCES BOOKS(ISBN);

ALTER TABLE BORROWS
ADD DAYSBORROWED NUMBER(10);

--This column shows how many days is/was the book borrowed
UPDATE BORROWS
SET DAYSBORROWED=(SELECT TO_DATE(BORROWS.RETURNDATE,'DD-MM-YYYY')-
TO_DATE (BORROWS.BORROWDATE,'DD-MM-YYYY')
FROM DUAL);

DESC BORROWS;
SELECT * FROM BORROWS;

```

2ο Μέρος:Εντολές δημιουργίας σχήματος

(functions_procedures_packages.sql file)

- Function BOOKPERCATEGORY

(Package:BOOKS_DB)

This function takes as an input parameter the category we want to search. It returns a cursor pointing to the isbn of every book that matches to this category.

```

--A FUNCTION FOR RETURNING ISBNs OF A CATEGORY
CREATE OR REPLACE PACKAGE TYPES AS
TYPE BOOK_CURSOR IS REF CURSOR;
END;
/

CREATE OR REPLACE FUNCTION BOOKPERCATEGORY(CAT IN BOOKS.CATEGORY%TYPE)
RETURN TYPES.BOOK_CURSOR
IS book TYPES.BOOK_CURSOR;

BEGIN
    OPEN book FOR
    SELECT ISBN
    FROM BOOKS WHERE CATEGORY=cat;
    RETURN book;
END BOOKPERCATEGORY;

```

```
/
SELECT BOOKPERCATEGORY('Art') FROM DUAL;
```

- Function CHECKAVALIABILITY

(Package:BORROWS_DB)

This function takes as an input parameter an isbn and returns if the book is available for borrowing or not. More especially, it returns a number 1 or 0. 1 represents that the book is available and 0 that it is not available.

```
CREATE OR REPLACE FUNCTION CHECKAVALIABILITY(ISBN_BOOK IN BOOKS.ISBN%TYPE)
RETURN NUMBER IS
check_avaliability NUMBER;

BEGIN
    SELECT AVALIABILITY INTO check_avaliability
    FROM BORROWS b JOIN BOOKS bo ON b.BOOKISBN=bo.ISBN
    WHERE ISBN=ISBN_BOOK;
    RETURN check_avaliability;
END;
/
```

Sql script for testing the function

```
DECLARE
    CHECK_AVALIABLE NUMBER;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Available');
    ELSE
    DBMS_OUTPUT.PUT_LINE('Not available');
    END IF;
END;
/

CHECK_AVALIABLE:=CHECKAVALIABILITY(22445566);
IF( CHECK_AVALIABLE = 1) THEN
```

- Function BORROWDATES

(Package:BORROWS_DB)

This function checks the table borrows; the column DAYSBORROWED of the input parameter ISBN_BOOK of a book and returns how many days was the book borrowed.

```

CREATE OR REPLACE FUNCTION BORROWDATES(ISBN_BOOK IN BOOKS.ISBN%TYPE)
RETURN NUMBER
IS DAYS_BORROWED NUMBER;

BEGIN
SELECT DAYSBORROWED INTO DAYS_BORROWED
    FROM BORROWS b JOIN BOOKS bo ON b.BOOKISBN=bo.ISBN
    WHERE ISBN=ISBN_BOOK;
    RETURN DAYS_BORROWED;
END;
/

```

Sql script for testing the function

```

DECLARE
    BORROWED_DAYS BORROWS.DAYSBORROWED%TYPE;
BEGIN
    BORROWED_DAYS:=BORROWDATES('35610290');
    DBMS_OUTPUT.PUT_LINE(BORROWED_DAYS);
END;
/

```

- Function PAYFINE

(Package: BORROWS_DB)

This function checks the value at the table borrows, at the column DAYSBORROWED at the line of the input parameter ISBN_BOOK. We assume that if the DAYSBORROWED is >30 the fine=10%*(DAYSBORROWED). This function returns the fine into the float variable fine.

```

CREATE OR REPLACE FUNCTION PAYFINE (ISBN_BOOK IN BOOKS.ISBN%TYPE)
RETURN FLOAT
IS FINE FLOAT;

DAYS_BORROWED NUMBER;
BEGIN
    SELECT DAYSBORROWED INTO DAYS_BORROWED
    FROM BORROWS b JOIN BOOKS bo ON b.BOOKISBN=bo.ISBN
    WHERE ISBN=ISBN_BOOK;

    IF (DAYS_BORROWED > 30) THEN
        FINE:=(10*DAYSBORROWED)/100;
    END IF;
    RETURN FINE;
END;/

```

Sql script for testing the function

```
DECLARE
    FINE FLOAT;
BEGIN
    FINE:=PAYFINE(31209765);
    DBMS_OUTPUT.PUT_LINE(FINE);
END;
/
```

- Procedure BOOKINFO

(Package:BOOKS_DB)

This procedure takes as an input parameter the title of a book and prints its category,ISBN and writer.

```
CREATE OR REPLACE PROCEDURE BOOKINFO (BOOK_TITLE IN BOOKS.TITLE%TYPE) AS
BEGIN
    DECLARE BOOK_ISBN BOOKS.ISBN%TYPE;
            BOOK_CATEGORY BOOKS.CATEGORY%TYPE;
            BOOK_WRITER BOOKS.WRITER%TYPE;
            CURSOR book IS
                SELECT ISBN,WRITER,CATEGORY FROM BOOKS WHERE TITLE=BOOK_TITLE;
    BEGIN
        OPEN book;
        LOOP
            FETCH book INTO BOOK_ISBN,BOOK_WRITER,BOOK_CATEGORY;
            EXIT WHEN book%NotFound;
        END LOOP;
        DBMS_OUTPUT.PUT_LINE('ISBN:' || BOOK_ISBN);
        DBMS_OUTPUT.PUT_LINE('TITLE:' || BOOK_TITLE);
        DBMS_OUTPUT.PUT_LINE('WRITER:' || BOOK_WRITER);
        DBMS_OUTPUT.PUT_LINE('CATEGORY:' || BOOK_CATEGORY);
        CLOSE book;

        END;
END;
/
```

```
EXEC BOOKINFO('Zazi in the train');
```

- Procedure BOOKINFO_W

(Package:BOOKS_DB)

This procedure takes as an input parameter the writer of a book and prints its category,ISBN and title.

```
CREATE OR REPLACE PROCEDURE BOOKINFO_W (BOOK_WRITER IN BOOKS.WRITER%TYPE)
AS
BEGIN
    DECLARE BOOK_ISBN BOOKS.ISBN%TYPE;
            BOOK_CATEGORY BOOKS.CATEGORY%TYPE;
            BOOK_TITLE BOOKS.TITLE%TYPE;
            CURSOR book IS
                SELECT ISBN,TITLE,CATEGORY FROM BOOKS WHERE
WRITER=BOOK_WRITER;
    BEGIN
    OPEN book;
    LOOP
        FETCH book INTO BOOK_ISBN,BOOK_TITLE,BOOK_CATEGORY;
        EXIT WHEN book%NotFound;
    END LOOP;
    DBMS_OUTPUT.PUT_LINE('ISBN:' || BOOK_ISBN);
    DBMS_OUTPUT.PUT_LINE('TITLE:' || BOOK_TITLE);
    DBMS_OUTPUT.PUT_LINE('WRITER:' || BOOK_WRITER);
    DBMS_OUTPUT.PUT_LINE('CATEGORY:' || BOOK_CATEGORY);

    CLOSE book;
    END;
END;
/
```

```
EXEC BOOKINFO_W('James Jous');
```

- Procedure BORROWBOOK

(Package:BORROS_DB)

This procedure takes as an input parameter the name of person who wants the book and the title of the book.Then checks if the book is available and if yes sets its member id with the new id of the member.It sets also the availability to 0 and also the borrowdate,returndate and daysborrowed columns from table BORROWS.

```
--A PROCEDURE TO BORROW A BOOK TO A MEMBER
CREATE OR REPLACE PROCEDURE BORROWBOOK(M_NAME IN
MEMBERS.NAME%TYPE,BOOK_TITLE IN BOOKS.TITLE%TYPE)
AS
```

```

BEGIN
  DECLARE
    BOOK_ISBN BOOKS.ISBN%TYPE;
    M_ID MEMBERS.ID%TYPE;
    BOOK_AVAIL BORROWS.AVALIABILLITY%TYPE;
    CURSOR book IS
      SELECT b.ISBN,bo.AVALIABILLITY
      FROM BOOKS b JOIN BORROWS bo
      ON b.ISBN=bo.BOOKISBN
      WHERE b.TITLE=BOOK_TITLE;

  BEGIN
    DBMS_OUTPUT.PUT_LINE('MEMBER:' || M_NAME || ' WANTS TO BORROW BOOK WITH
TITLE:' || BOOK_TITLE);
    OPEN book;
    LOOP
      FETCH book INTO BOOK_ISBN,BOOK_AVAIL;
      EXIT WHEN book%NotFound;
    END LOOP;
    SELECT ID INTO M_ID FROM MEMBERS WHERE NAME=M_NAME;
    --DBMS_OUTPUT.PUT_LINE(BOOK_AVAIL || BOOK_ISBN);
    IF (BOOK_AVAIL=1) THEN
      UPDATE BORROWS
      SET MEMBERID=M_ID WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET AVALIABILLITY=0 WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET BORROWDATE=sysdate WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET RETURNDATE=NULL WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET DAYSBORROWED=NULL WHERE BOOKISBN=BOOK_ISBN;

      DBMS_OUTPUT.PUT_LINE('DONE!');

    ELSE
      DBMS_OUTPUT.PUT_LINE('The book is not available');
    END IF;

    CLOSE book;
  END;
END;
/

```

```
EXEC BORROWBOOK('Maria Kitsou','Olive again');
EXEC BORROWBOOK('John Doe','Me and you');
EXEC BORROWBOOK('Jane Eyer','Zazi in the train');
```

- Procedure RETURNBORROWEDBOOK

(Package: BORROWS_DB)

This procedure takes as an input parameter the title of a book we want to return. It checks firstly, if the book is borrowed or not. If it is borrowed, the book must be returned. So, the availability=1 (available for borrowing), returndate=sysdate, memberid=null, and we update also the daysborrowed (from borrowdate till sysdate). If it is available we simply print that the book is available.

```
--A PROCEDURE TO RETURN A BOOK WHICH IS IN BORROW MODE
CREATE OR REPLACE PROCEDURE RETURNBORROWEDBOOK(BOOK_TITLE IN
BOOKS.TITLE%TYPE)
AS
BEGIN
  DECLARE
    BOOK_ISBN BOOKS.ISBN%TYPE;
    BOOK_AVAIL BORROWS.AVALIABILLITY%TYPE;
    CURSOR book IS
      SELECT b.ISBN,bo.AVALIABILLITY
      FROM BOOKS b JOIN BORROWS bo
      ON b.ISBN=bo.BOOKISBN
      WHERE b.TITLE=BOOK_TITLE;

  BEGIN
    OPEN book;
    LOOP
      FETCH book INTO BOOK_ISBN,BOOK_AVAIL;
      EXIT WHEN book%NotFound;
    END LOOP;
    --DBMS_OUTPUT.PUT_LINE(BOOK_ISBN || BOOK_AVAIL);
    IF (BOOK_AVAIL=0) THEN
      DBMS_OUTPUT.PUT_LINE('Book must be returned');
      UPDATE BORROWS
      SET AVALIABILLITY=1 WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET RETURNDATE=SYSDATE WHERE BOOKISBN=BOOK_ISBN;

      UPDATE BORROWS
      SET MEMBERID=null WHERE BOOKISBN=BOOK_ISBN;
```

```

        UPDATE BORROWS
        SET DAYSBORROWED=(SELECT
TO_DATE(BORROWS.RETURNDATE,'DD-MM-YYYY')-
        TO_DATE (BORROWS.BORROWDATE,'DD-MM-YYYY')
        FROM DUAL);

        ELSE
        DBMS_OUTPUT.PUT_LINE('Book is available');
        END IF;
    END;
END;
/

```

```

EXEC RETURNBORROWEDBOOK('Me and You');
EXEC RETURNBORROWEDBOOK('Zazi in the train');

```

- PROCEDURE INSERTBOOK

(Package:BOOKS_DB)

This procedure takes as input parameters the title,the isbn,the author and the category of the book we want to insert.

```

CREATE OR REPLACE PROCEDURE INSERTBOOK(BOOK_TITLE IN BOOKS.TITLE%TYPE,
                                         BOOK_WRITER IN BOOKS.WRITER%TYPE,
                                         BOOK_ISBN IN BOOKS.ISBN%TYPE,
                                         BOOK_CATEGORY IN
BOOKS.CATEGORY%TYPE
                                         )
AS
    BEGIN
        INSERT INTO
BOOKS(TITLE,WRITER,ISBN,CATEGORY)VALUES(BOOK_TITLE,BOOK_WRITER,BOOK_ISBN,BOOK_CATEGORY);
    END INSERTBOOK;
/

```

```

EXEC INSERTBOOK('Flower','Sabine Bont',09872345,'Tales');

```

- Procedure INSERTMEMEBER

(Package:MEMBERS DB)

This procedure takes as input parameters the member's id,the member's name and the member's address and insert this member at the table members.The id is created as auto-generate in the table of members.

```
--A PROCEDURE TO INSERT A NEW MEMBER
CREATE OR REPLACE PROCEDURE INSERTMEMBER(M_ID IN MEMBERS.ID%TYPE,M_NAME IN
MEMBERS.NAME%TYPE,M_ADDRESS IN MEMBERS.ADDRESS%TYPE)
AS
    BEGIN
        INSERT INTO MEMBERS(ID,NAME,ADDRESS)VALUES(M_ID,M_NAME,M_ADDRESS);
    END;
/
```

```
EXEC INSERTMEMBER(members_seq.nextval,'George Grigoriou','80
Selimon,Equador,BR');
```

- Procedure UPDATERETURNDATE

(Package:BORROWS DB)

This procedure takes as input parameter the title of the book which must be returned and the return date.First it checks if the book is available.If yes,means that the book is not borrowed.If the book is not available,means still borrowed.So,we set avaliability=1(we assume that after returning book, is available)memberid=null,returndate=the input parameter return_date and we compute again how many days was the book borrowed.

```
--A PROCEDURE TO UPDATE THE RETURN DATE IN A STILL BORROWED BOOK
CREATE OR REPLACE PROCEDURE UPDATERETURNDATE(BOOK_ISBN IN
BOOKS.ISBN%TYPE,RETURN_DATE IN BORROWS.RETURNDATE%TYPE)
AS
    BEGIN
        DECLARE
            BOOK_AVAIL BORROWS.AVALIABILLITY%TYPE;
            CURSOR book IS
                SELECT bo.AVALIABILLITY
                FROM BOOKS b JOIN BORROWS bo
                ON b.ISBN=bo.BOOKISBN
                WHERE b.ISBN=BOOK_ISBN;
        BEGIN
            OPEN book;
            LOOP
```

```

    FETCH book INTO BOOK_AVAIL;
    EXIT WHEN book%NotFound;
END LOOP;

IF(BOOK_AVAIL=1) THEN
    DBMS_OUTPUT.PUT_LINE('Book is still available');
ELSE
    DBMS_OUTPUT.PUT_LINE('Book is still borrowed');
    UPDATE BORROWS
    SET RETURNDATE=RETURN_DATE WHERE BOOKISBN=BOOK_ISBN;

    UPDATE BORROWS
    SET AVAILABILITY=1 WHERE BOOKISBN=BOOK_ISBN;

    UPDATE BORROWS
    SET MEMBERID=NULL WHERE BOOKISBN=BOOK_ISBN;

    UPDATE BORROWS
    SET DAYSBORROWED=(SELECT TO_DATE(BORROWS.RETURNDATE, 'DD-MM-YYYY')-
    TO_DATE (BORROWS.BORROWDATE, 'DD-MM-YYYY')
    FROM DUAL);
END IF;
END;

END;
/

```

```
EXEC UPDATERETURNDATE(1563221,'15-08-2020');
EXEC UPDATERETURNDATE(91256109,'15-08-2020');
```

3ο Μέρος: Δημιουργία πακέτων PL/SQL

(functions_procedures_packages.sql file)

- Package:BOOK DB

```
--PACKAGE BOOK_DB
CREATE OR REPLACE PACKAGE BOOK_DB AS
TYPE book_cur IS REF CURSOR;
FUNCTION BOOKPERCATEGORY(CAT BOOKS.CATEGORY%TYPE) RETURN book_cur;
PROCEDURE BOOKINFO (BOOK_TITLE IN BOOKS.TITLE%TYPE);
PROCEDURE BOOKINFO_W (BOOK_WRITER IN BOOKS.WRITER%TYPE);
PROCEDURE INSERTBOOK(BOOK_TITLE IN BOOKS.TITLE%TYPE,
                     BOOK_WRITER IN BOOKS.WRITER%TYPE,
```

```
        BOOK_ISBN IN BOOKS.ISBN%TYPE,  
        BOOK_CATEGORY IN BOOKS.CATEGORY%TYPE);  
END BOOK_DB;  
/
```

- Package:MEMBERS_DB

```
--PACKAGE MEMBERS_DB  
CREATE OR REPLACE PACKAGE MEMBERS_DB AS  
    PROCEDURE INSERTMEMBER(M_ID IN MEMBERS.ID%TYPE,M_NAME IN  
MEMBERS.NAME%TYPE,M_ADDRESS IN MEMBERS.ADDRESS%TYPE);  
END MEMBERS_DB;  
/
```

- Package:BORROWS_DB

```
--PACKAGE BORROWS_DB  
CREATE OR REPLACE PACKAGE BORROWS_DB AS  
    FUNCTION CHECKAVALIABILITY(ISBN_BOOK IN BOOKS.ISBN%TYPE)RETURN NUMBER;  
    FUNCTION BORROWDATES(ISBN_BOOK IN BOOKS.ISBN%TYPE) RETURN NUMBER;  
    FUNCTION PAYFINE (ISBN_BOOK IN BOOKS.ISBN%TYPE) RETURN FLOAT;  
    PROCEDURE BORROWBOOK(M_NAME IN MEMBERS.NAME%TYPE,BOOK_TITLE IN  
BOOKS.TITLE%TYPE);  
    PROCEDURE RETURNBORROWEDBOOK(BOOK_TITLE IN BOOKS.TITLE%TYPE);  
    PROCEDURE UPDATERETURNDATE(BOOK_ISBN IN BOOKS.ISBN%TYPE,RETURN_DATE IN  
BORROWS.RETURNDATE%TYPE);
```